Lab no. 5

Build, Test, and Package Applications with IntelliJ IDEA and Docker

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Task 1: Clone and Launch IntelliJ IDEA

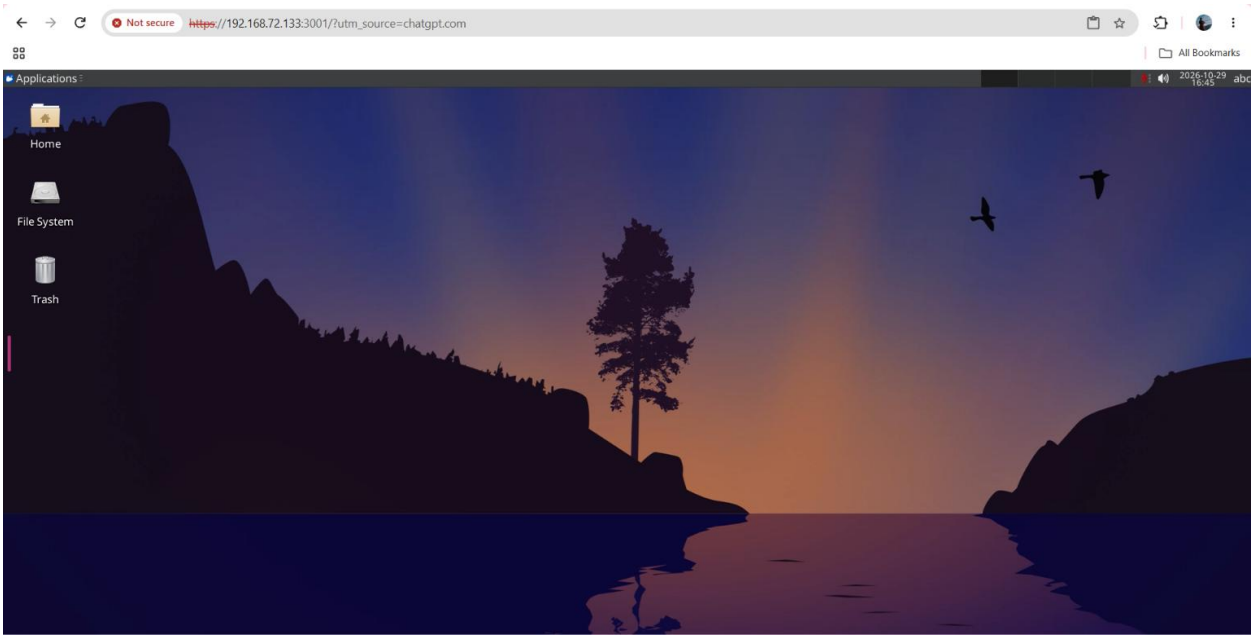
```
sania270804@ubuntu:~$ cd ~
git clone https://github.com/WaqasSaleem97/Intellij-Idea-docker.git
cd Intellij-Idea-docker
Cloning into 'Intellij-Idea-docker'...
remote: Enumerating objects: 35, done.
remote: Counting objects: 100% (35/35), done.
remote: Compressing objects: 100% (32/32), done.
remote: Total 35 (delta 16), reused 5 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (35/35), 15.24 KiB | 123.00 KiB/s, done.
Resolving deltas: 100% (16/16), done.
sania270804@ubuntu:~/Intellij-Idea-docker$
```

In your Ubuntu host terminal, clone the supplied repository.

```
sania270804@ubuntu:~/Intellij-Idea-docker$ bash setup.sh
Using your existing .env configuration.
[+] Building 1001.6s (12/12) FINISHED
 => [internal] load local bake definitions                                                                                                0.0s
 => => reading from stdin 642B                                                                                                            0.0s
 => [internal] load build definition from Dockerfile                                                                                      0.1s
 => => transferring dockerfile: 1.47kB                                                                                                    0.0s
 => [internal] load metadata for lscr.io/linuxserver/webtop:ubuntu-xfce                                                                   2.4s
 => [internal] load .dockerignore                                                                                                         0.0s
 => => transferring context: 95B                                                                                                          0.0s
 => [1/5] FROM lscr.io/linuxserver/webtop:ubuntu-xfce@sha256:432475b977d9ff06f8f8adfb9608541bbde51a566ac0cbe62af44f33265e1a37             0.0s
 => => resolve lscr.io/linuxserver/webtop:ubuntu-xfce@sha256:432475b977d9ff06f8f8adfb9608541bbde51a566ac0cbe62af44f33265e1a37             0.1s
 => [internal] load build context                                                                                                         0.1s
 => => transferring context: 1.66kB                                                                                                       0.0s
 => [2/5] RUN apt-get update     && DEBIAN_FRONTEND=noninteractive apt-get install -y --no-install-recommends        ca-certificates curl git openjd  97.9s
 => [3/5] RUN set -eu;     case "$(dpkg --print-architecture)" in       arm64) archive="idea-2026.2.3-aarch64.tar.gz" ;;       amd64) archive="idea  649.1s
 => [4/5] COPY intellij-idea.desktop /usr/share/applications/intellij-idea.desktop                                                        0.1s
 => [5/5] COPY --chmod=0755 init-terminal.sh /custom-cont-init.d/10-terminal-prompt                                                       0.1s
 => exporting to image                                                                                                                  249.0s
 => => exporting layers                                                                                                                 204.4s
 => => exporting manifest sha256:f82fec9e62eacfa954d4bdd77d6fde2e9d582c2ed79e932803ef110f2ccbc472                                          0.0s
 => => exporting config sha256:a8e2ad05e250c371e54079f42df010577f4e41fac9fa691c6e940f5af909bbb580                                          0.0s
 => => exporting attestation manifest sha256:f77546361cb2bb4726d4539edf85cdb5600fcbcf8d09bb839cd699113c8d1fbf                              0.0s
 => => exporting manifest list sha256:b25be5fea7d78d001401c84d6c1bdbaa96c9e686d2a4c0b99fd53d31dab42ca9                                     0.0s
 => => naming to docker.io/library/intellij-idea-docker-intellij:latest                                                                   0.0s
 => => unpacking to docker.io/library/intellij-idea-docker-intellij:latest                                                                45.3s
 => resolving provenance for metadata file                                                                                                0.0s
[+] build 1/1
 ✔ Image intellij-idea-docker-intellij Built                                                                                           1001.8s
[+] up 2/2
 ✔ Network intellij-idea-docker_default      Created                                                                                      0.1s
 ✔ Container intellij-idea-docker-intellij-1 Started                                                                                      1.7s
NAME                          IMAGE                          COMMAND   SERVICE    CREATED         STATUS                  PORTS
intellij-idea-docker-intellij-1   intellij-idea-docker-intellij   "/init"   intellij   2 seconds ago   Up Less than a second   3000/tcp, 127.0.0.1:3001->3001/tcp

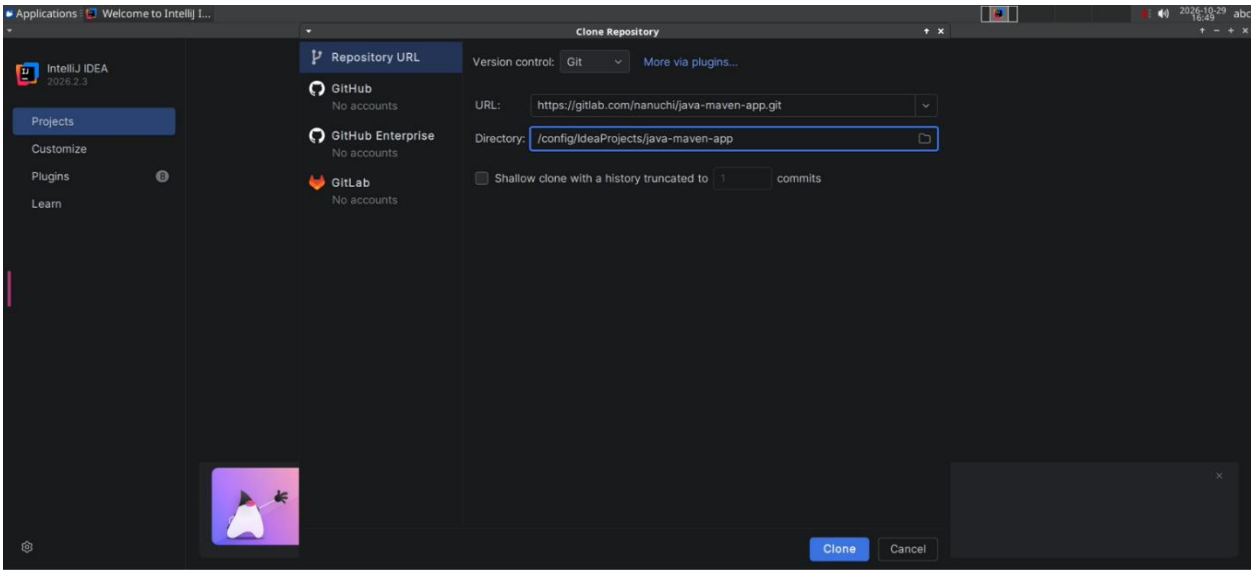
Browser username: admin (or WEBTOP_USER from your existing .env).
Open https://localhost:3001 in your host computer browser.
sania270804@ubuntu:~/Intellij-Idea-docker$ docker compose ps
NAME                          IMAGE                          COMMAND   SERVICE    CREATED              STATUS          PORTS
intellij-idea-docker-intellij-1   intellij-idea-docker-intellij   "/init"   intellij   About a minute ago   Up 58 seconds   3000/tcp, 127.0.0.1:3001->3001/tcp
sania270804@ubuntu:~/Intellij-Idea-docker$
```

Build and launch the environment



Open https://localhost:3001 in your host computer's browser.

Task 2: Build and Run the Maven Project



On IntelliJ's Welcome screen, select Clone Repository.

```
sania270804@ubuntu:~/IdeaProjects/java-maven-app$ mvn --version
Apache Maven 3.9.12
Maven home: /usr/share/maven
Java version: 21.0.12.1, vendor: Ubuntu, runtime: /usr/lib/jvm/java-21-openjdk-amd64
Default locale: en_US, platform encoding: UTF-8
OS name: "linux", version: "7.0.0-38-generic", arch: "amd64", family: "unix"

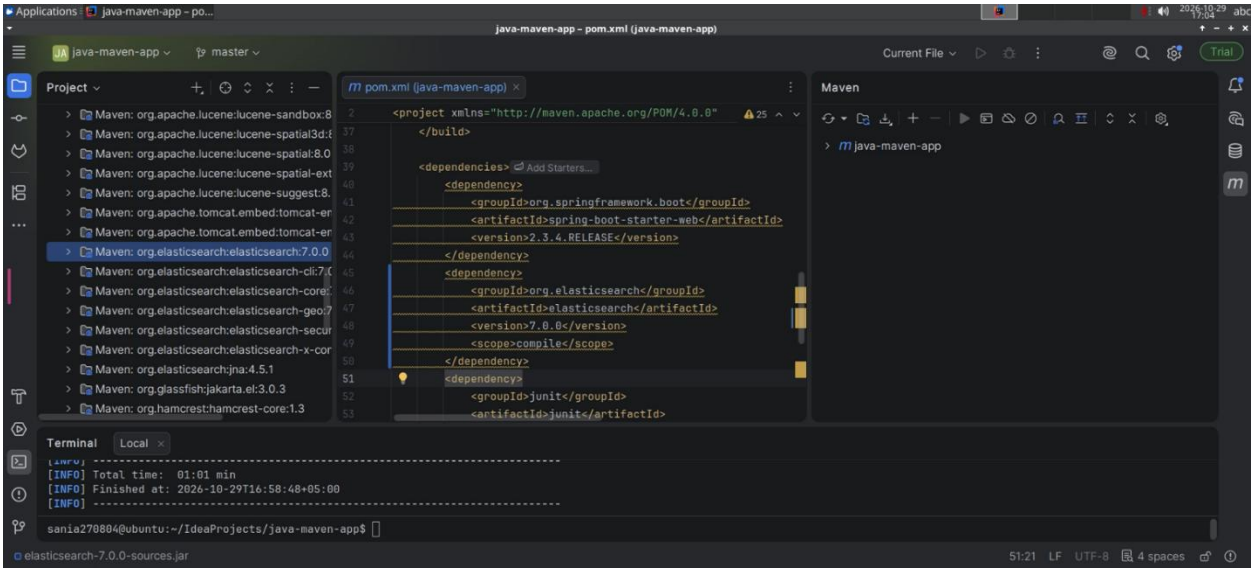
sania270804@ubuntu:~/IdeaProjects/java-maven-app$
```

Open View → Tool Windows → Terminal inside IntelliJ. Work from the Maven project's root directory.

```
[INFO] Replacing main artifact with repackaged archive
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  01:01 min
[INFO] Finished at: 2026-10-29T16:58:48+05:00
[INFO] ------------------------------------------------------------------------

sania270804@ubuntu:~/IdeaProjects/java-maven-app$
```

Package the application.



Open pom.xml in IntelliJ. Add this dependency inside the existing <dependencies> section, alongside the other project dependencies.

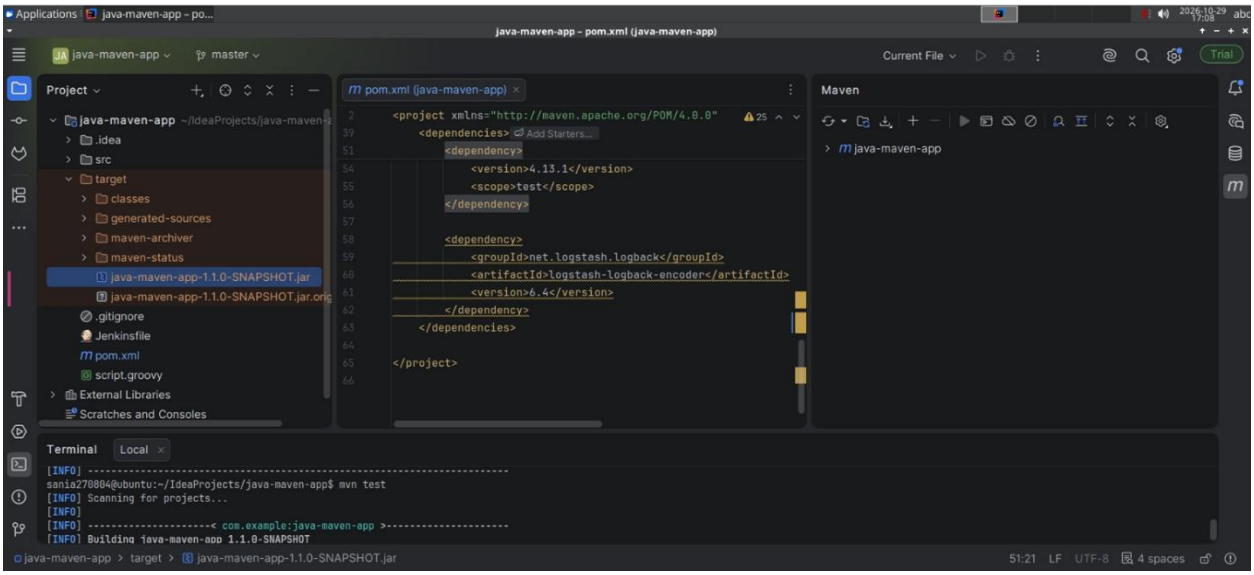
```
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /config/IdeaProjects/java-maven-app/src/test/resources
[INFO]
[INFO] --- compiler:3.6.0:testCompile (default-testCompile) @ java-maven-app ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:2.17:test (default-test) @ java-maven-app ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.1.2:jar (default-jar) @ java-maven-app ---
[INFO] Building jar: /config/IdeaProjects/java-maven-app/target/java-maven-app-1.1.0-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:2.3.5.RELEASE:repackage (default) @ java-maven-app ---
[INFO] Replacing main artifact with repackaged archive
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  13.176 s
[INFO] Finished at: 2026-10-29T17:05:07+05:00
[INFO] ------------------------------------------------------------------------

sania270804@ubuntu:~/IdeaProjects/java-maven-app$
```

Package the project again after adding the dependency.

```
sania270804@ubuntu:~/IdeaProjects/java-maven-app$ mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] ---------------------< com.example:java-maven-app >---------------------
[INFO] Building java-maven-app 1.1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.6.0:compile (default-compile) @ java-maven-app ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /config/IdeaProjects/java-maven-app/src/test/resources
[INFO]
[INFO] --- compiler:3.6.0:testCompile (default-testCompile) @ java-maven-app ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:2.17:test (default-test) @ java-maven-app ---
[INFO] No tests to run.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.631 s
[INFO] Finished at: 2026-10-29T17:05:45+05:00
```

Run Maven's test phase.



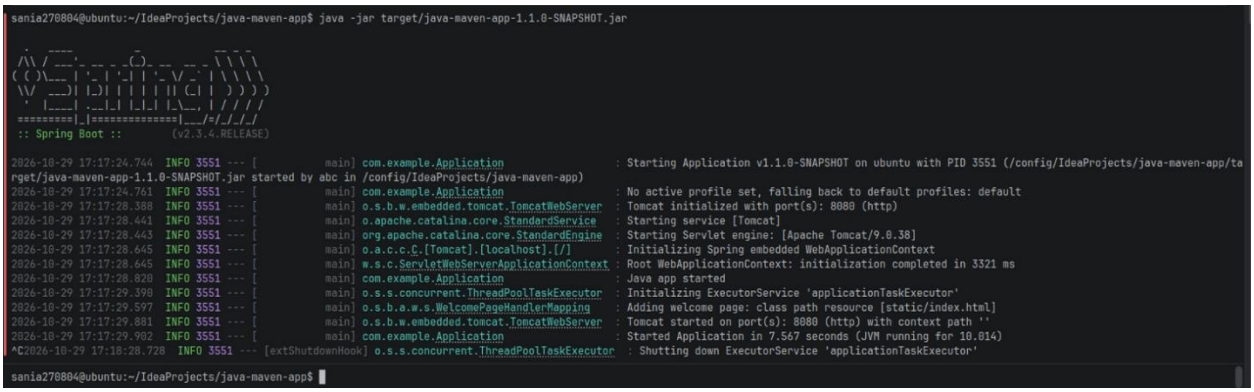
In IntelliJ's Project pane, refresh the project if needed and expand target.

```
sania270804@ubuntu:~/IdeaProjects/java-maven-app$ java -jar target/java-maven-app-1.1.0-SNAPSHOT.jar

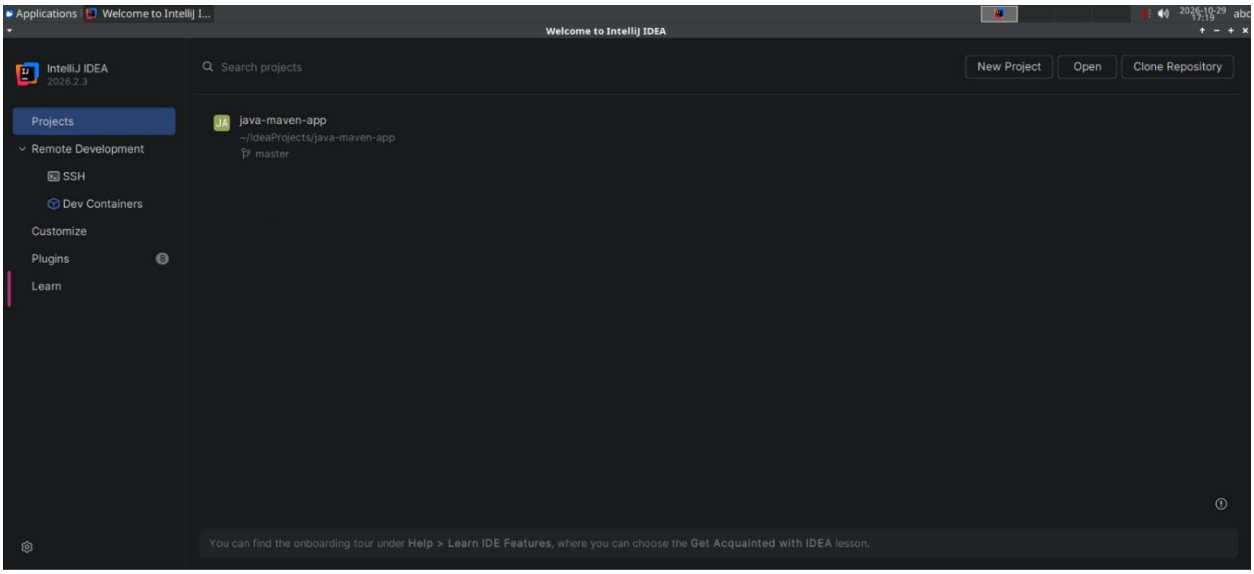
  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/
 :: Spring Boot ::                (v2.3.4.RELEASE)

2026-10-29 17:17:24.744  INFO 3551 --- [           main] com.example.Application                  : Starting Application v1.1.0-SNAPSHOT on ubuntu with PID 3551 (/config/IdeaProjects/java-maven-app/target/java-maven-app-1.1.0-SNAPSHOT.jar started by abc in /config/IdeaProjects/java-maven-app)
2026-10-29 17:17:24.761  INFO 3551 --- [           main] com.example.Application                  : No active profile set, falling back to default profiles: default
2026-10-29 17:17:28.388  INFO 3551 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat initialized with port(s): 8080 (http)
2026-10-29 17:17:28.441  INFO 3551 --- [           main] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-10-29 17:17:28.443  INFO 3551 --- [           main] org.apache.catalina.core.StandardEngine  : Starting Servlet engine: [Apache Tomcat/9.0.38]
2026-10-29 17:17:28.645  INFO 3551 --- [           main] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing Spring embedded WebApplicationContext
2026-10-29 17:17:28.645  INFO 3551 --- [           main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 3321 ms
2026-10-29 17:17:28.828  INFO 3551 --- [           main] com.example.Application                  : Java app started
2026-10-29 17:17:29.398  INFO 3551 --- [           main] o.s.s.concurrent.ThreadPoolTaskExecutor  : Initializing ExecutorService 'applicationTaskExecutor'
2026-10-29 17:17:29.597  INFO 3551 --- [           main] o.s.b.a.w.s.WelcomePageHandlerMapping    : Adding welcome page: class path resource [static/index.html]
2026-10-29 17:17:29.881  INFO 3551 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat started on port(s): 8080 (http) with context path ''
2026-10-29 17:17:29.902  INFO 3551 --- [           main] com.example.Application                  : Started Application in 7.567 seconds (JVM running for 10.014)
```

In the same IntelliJ terminal, run the JAR from the project root

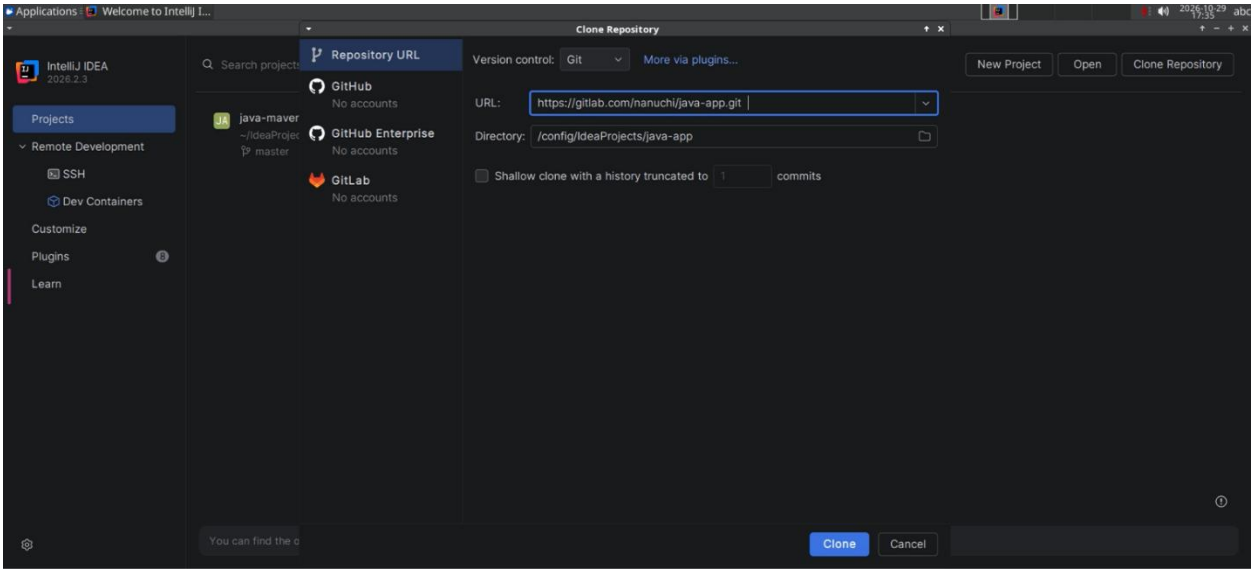


Press Ctrl+C in that terminal to stop the application. Wait for the shell prompt to return.



Select File → Close Project and return to IntelliJ's Welcome screen.

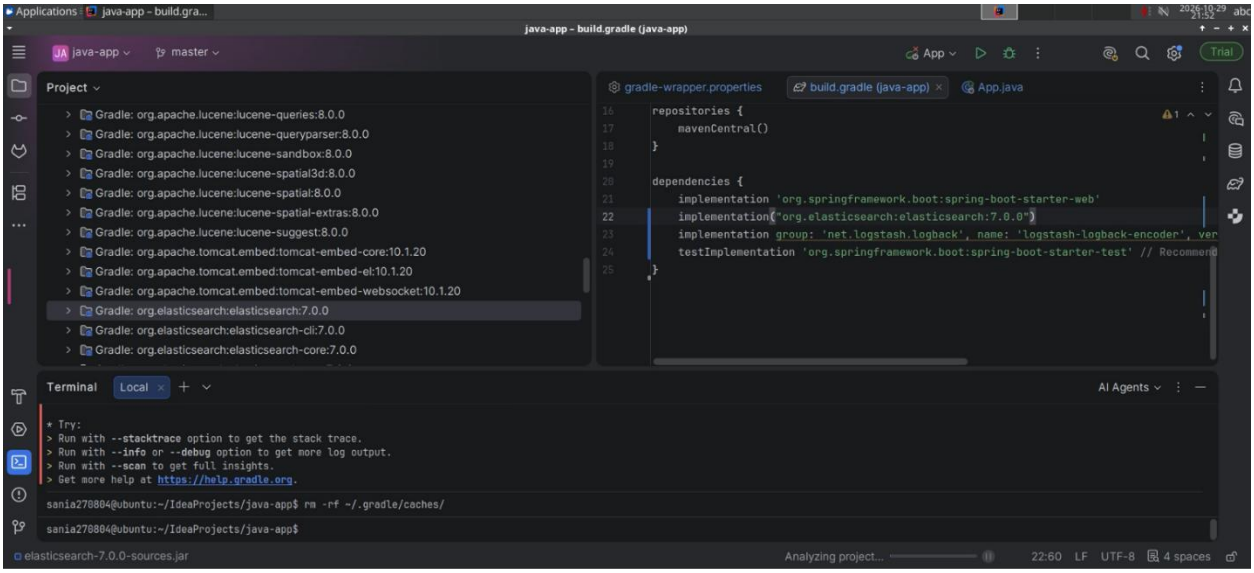
Task 3: Build and Run the Gradle Project



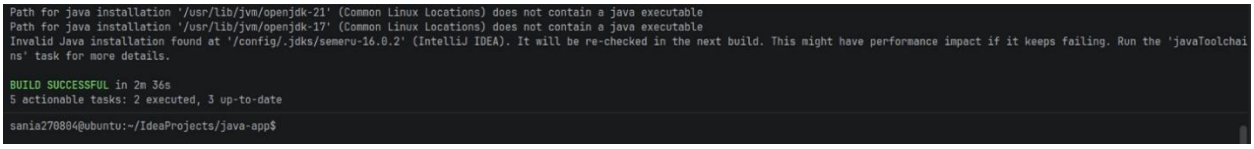
From IntelliJ's Welcome screen, use Clone Repository



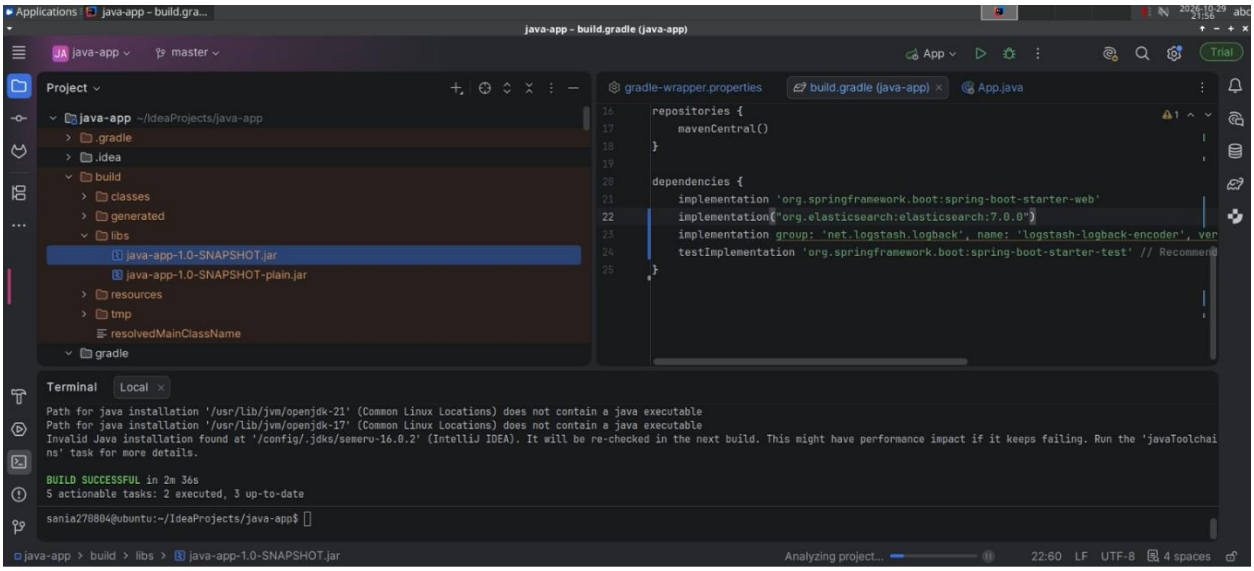
Open View → Tool Windows → Terminal. The wrapper downloads the project's Gradle version on its first run. Wait for BUILD SUCCESSFUL.



Open build.gradle and add this line inside the existing dependencies { ... } block.



Build the project again after the dependency change



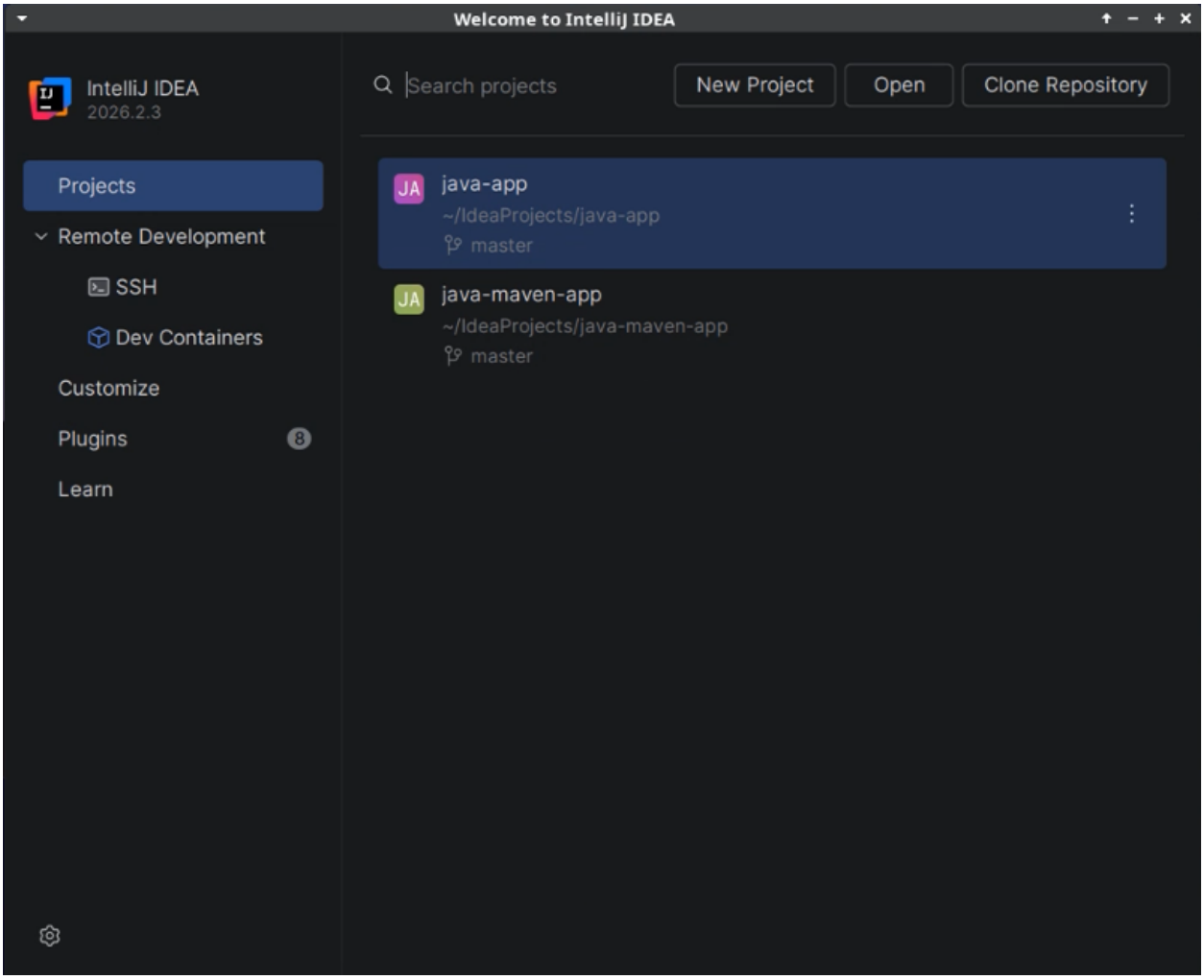
In IntelliJ's Project pane, expand build → libs.



Run the generated JAR from the project root.

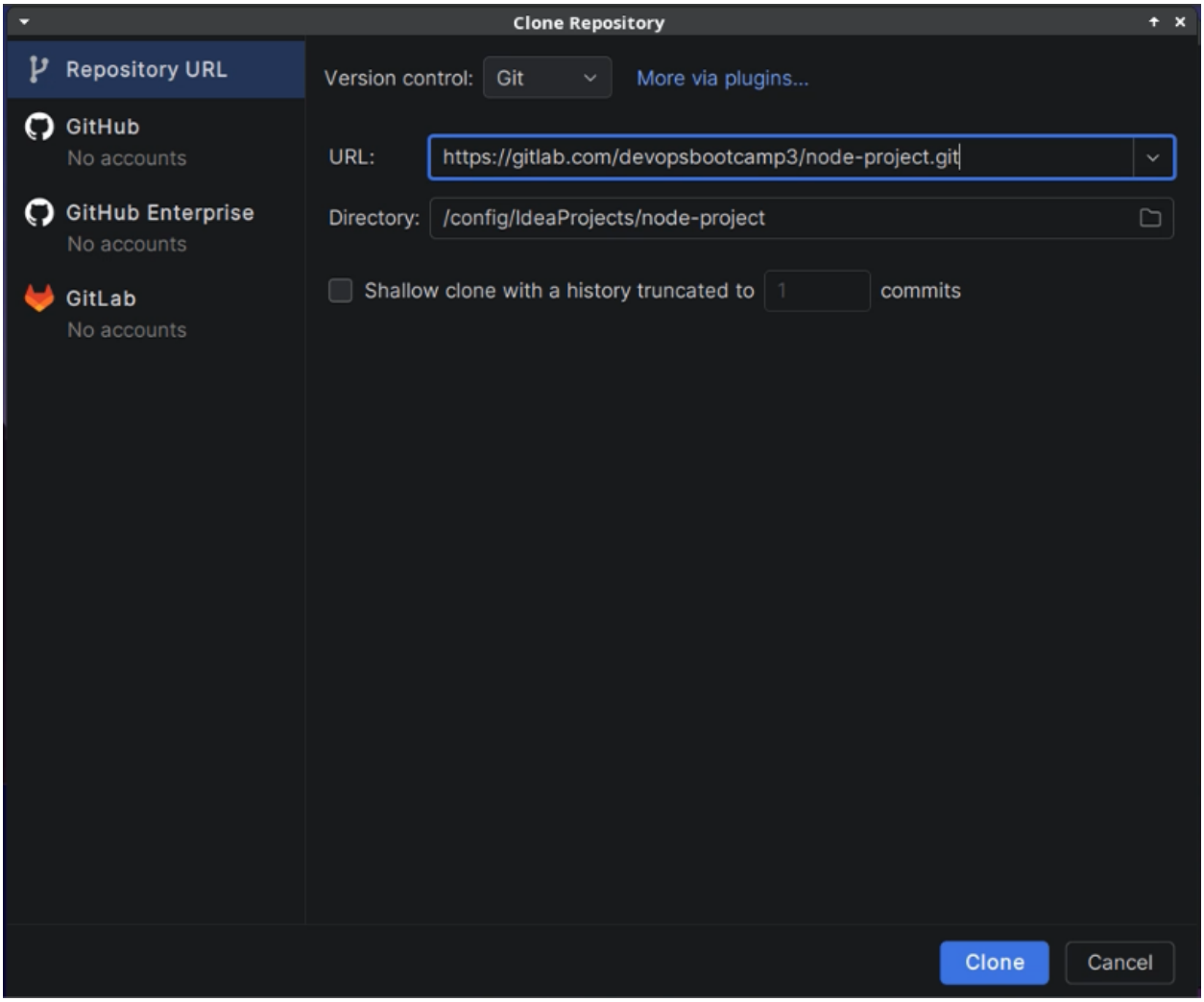


Press Ctrl+C and wait for the prompt to return. This also releases port 8080 before another application uses it.



Select File → Close Project.

Task 4: Install and Run the Node.js Project



Use IntelliJ's Clone Repository / Get from Version Control action.

```
sania270804@ubuntu:~/IdeaProjects/node-project$ node --version
v22.23.4

sania270804@ubuntu:~/IdeaProjects/node-project$ npm --version
10.9.9

sania270804@ubuntu:~/IdeaProjects/node-project$
```

Open the terminal inside IntelliJ and install Node.js together with npm.

```
npm warn deprecated is-accessor-descriptor@0.1.6: Please upgrade to v0.1.7
npm warn deprecated is-accessor-descriptor@0.1.6: Please upgrade to v0.1.7
npm warn deprecated is-data-descriptor@0.1.4: Please upgrade to v0.1.5
npm warn deprecated is-data-descriptor@0.1.4: Please upgrade to v0.1.5
npm warn deprecated is-accessor-descriptor@0.1.6: Please upgrade to v0.1.7
npm warn deprecated is-accessor-descriptor@0.1.6: Please upgrade to v0.1.7
npm warn deprecated uuid@8.3.2: uuid@10 and below is no longer supported.  For ESM codebases, update to uuid@latest.  For CommonJS codebases, use uuid@11 (but be aware this version will likely be depr
ecated in 2028).

added 554 packages, and audited 555 packages in 3m

24 packages are looking for funding
  run `npm fund` for details

54 vulnerabilities (10 low, 21 moderate, 19 high, 4 critical)

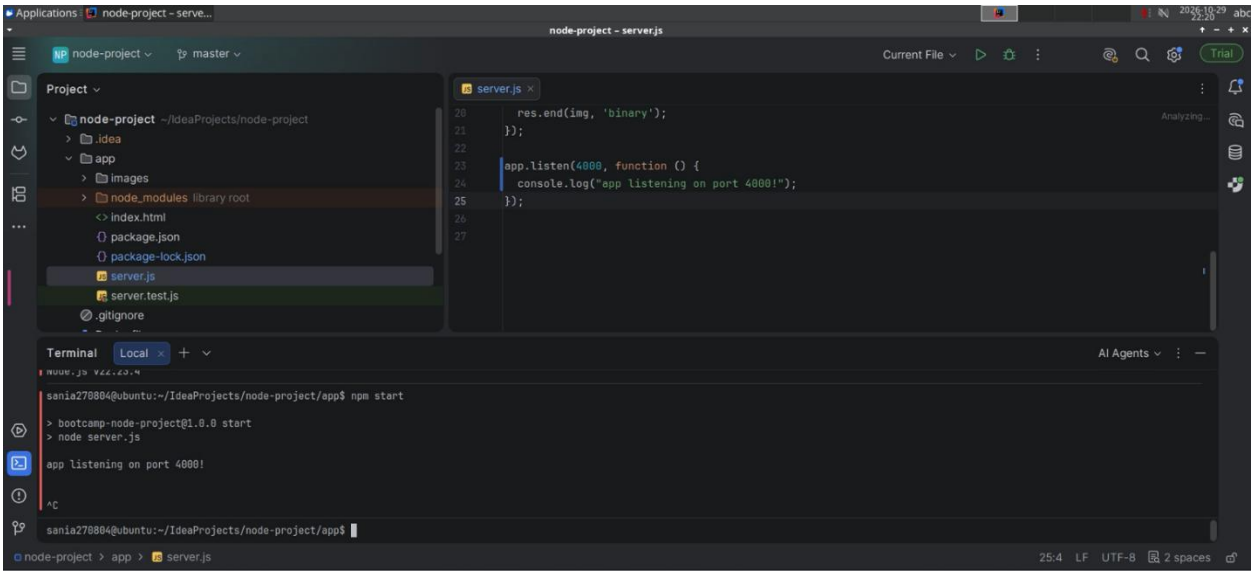
To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

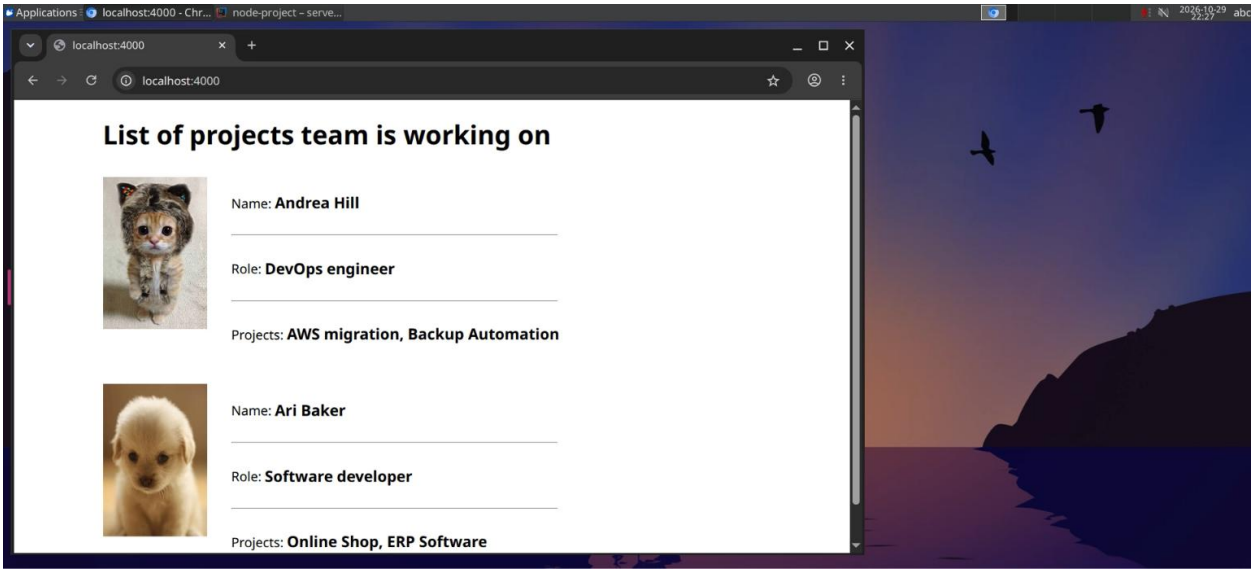
Run `npm audit` for details.
npm notice
npm notice New major version of npm available! 10.9.9 -> 12.2.1
npm notice Changelog: https://github.com/npm/cli/releases/tag/v12.2.1
npm notice To update run: npm install -g npm@12.2.1
npm notice

sania270804@ubuntu:~/IdeaProjects/node-project/app$
```

From the project root, enter app and install its dependencies.



Start the application.



Keep the application running. Open a browser inside the container's desktop from its applications menu and visit http://localhost:4000. Verify that the Node.js project page and its profile images are visible.

```
sania270804@ubuntu:~/IdeaProjects/node-project/app$ npm test

> bootcamp-node-project@1.0.0 test
> jest

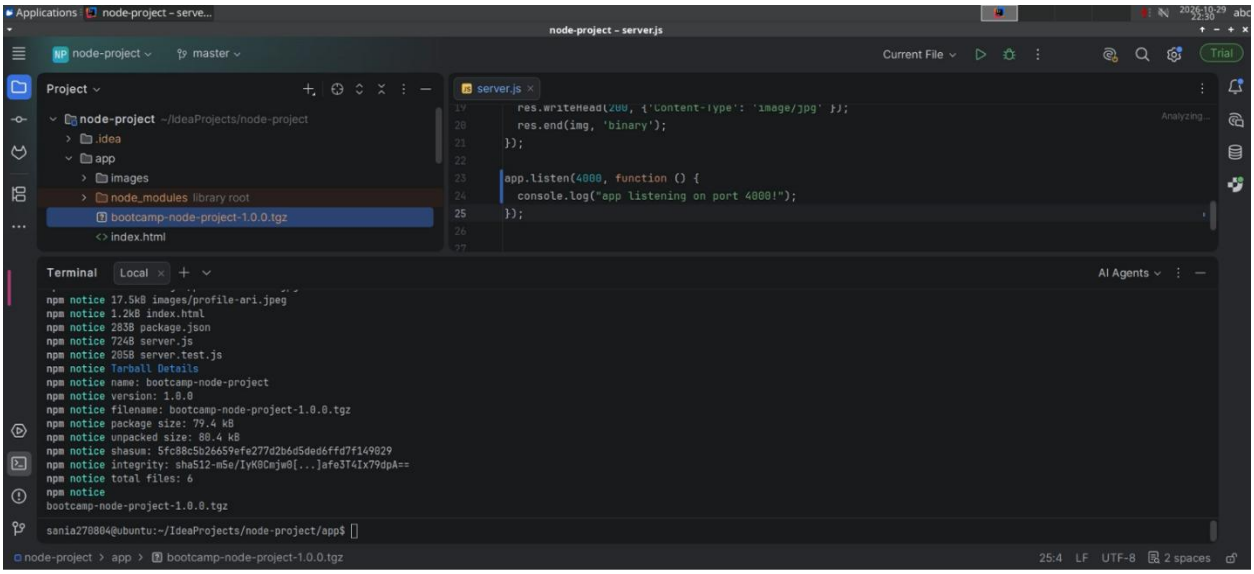
Browserslist: caniuse-lite is outdated. Please run:
npx browserslist@latest --update-db

Why you should do it regularly:
https://github.com/browserslist/browserslist#browsers-data-updating
PASS ./server.test.js
  ✓ main index.html file exists (7 ms)

Test Suites: 1 passed, 1 total
Tests:       1 passed, 1 total
Snapshots:   0 total
Time:        8.509 s
Ran all test suites.

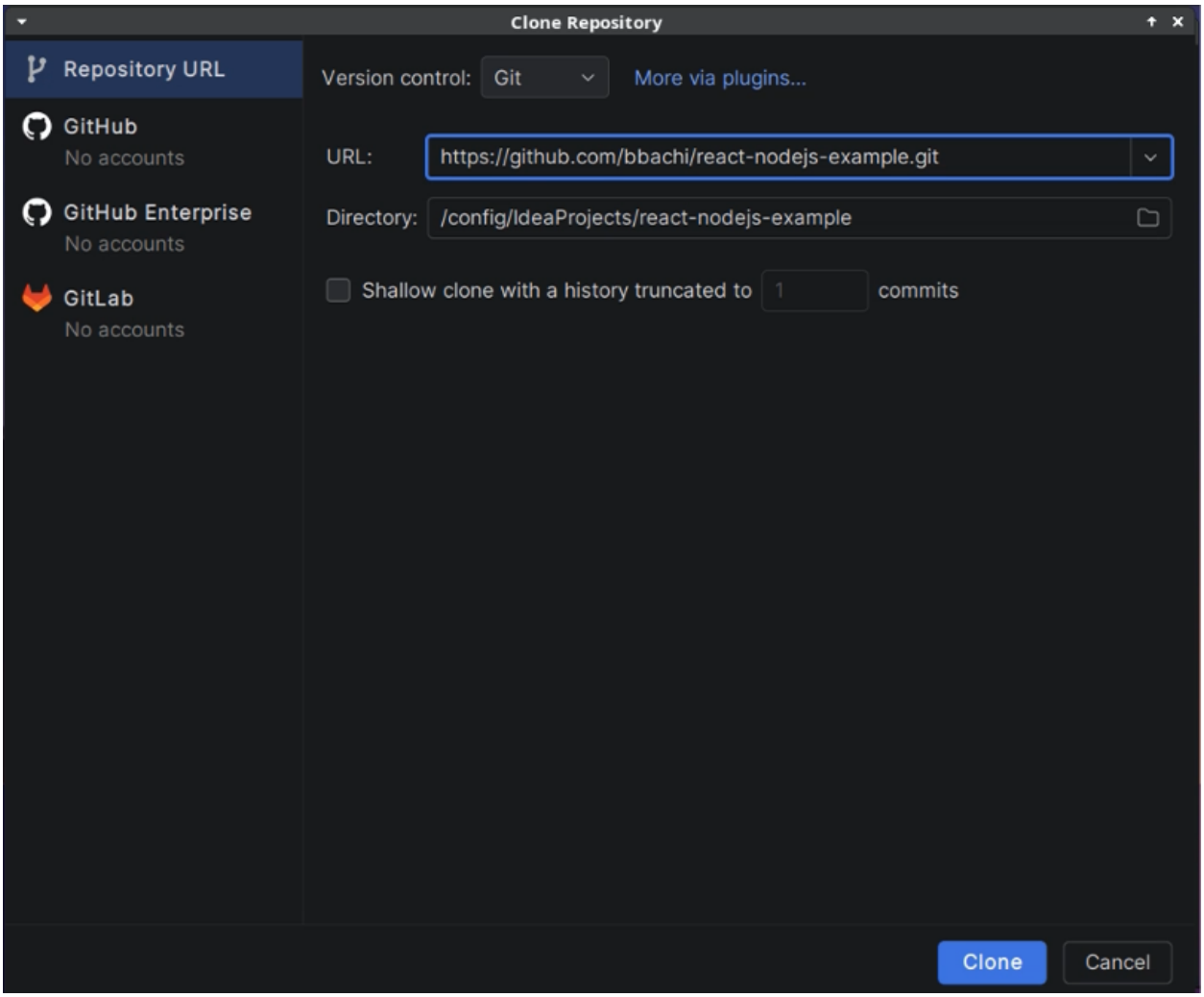
sania270804@ubuntu:~/IdeaProjects/node-project/app$
```

Stay in the app directory and run the application's tests



Create the application's package artifact.

Task 5: Build the React and Node.js API Artifact



From IntelliJ's Welcome screen, use Clone Repository.

```
sania270804@ubuntu:~/IdeaProjects/react-nodejs-example/api$ npm install
npm warn old lockfile
npm warn old lockfile The package-lock.json file was created with an old version of npm,
npm warn old lockfile so supplemental metadata must be fetched from the registry.
npm warn old lockfile
npm warn old lockfile This is a one-time fix-up, please be patient...
npm warn deprecated rimraf@2.7.1: Rimraf versions prior to v4 are no longer supported
npm warn deprecated rimraf@3.0.2: Rimraf versions prior to v4 are no longer supported

changed 48 packages, and audited 754 packages in 3m

15 packages are looking for funding
  run `npm fund` for details

64 vulnerabilities (5 low, 17 moderate, 35 high, 7 critical)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
sania270804@ubuntu:~/IdeaProjects/react-nodejs-example/api$
```

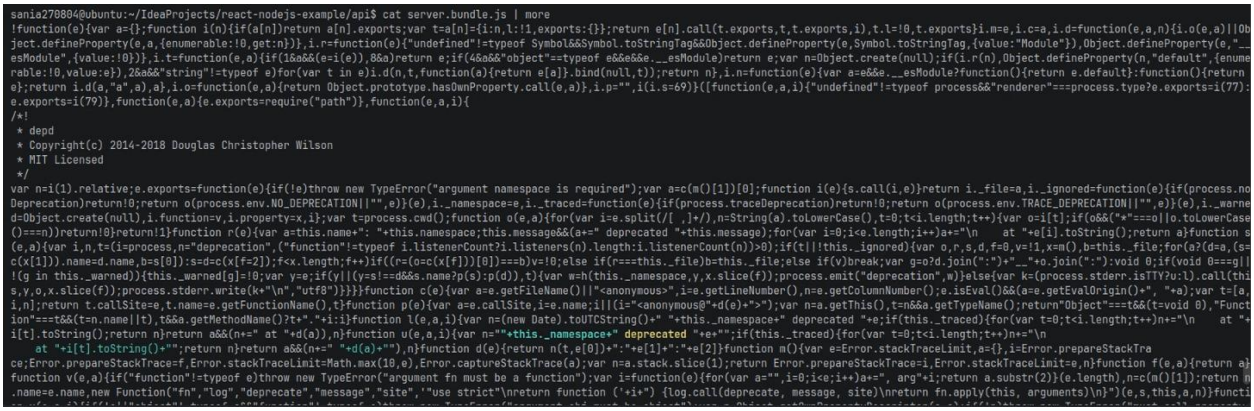
After installation, refresh the Project pane if needed and expand api. Locate node_modules, which contains the installed dependencies.

```
Time: 10162ms
Built at: 10/29/2026 11:06:32 PM
          Asset     Size  Chunks             Chunk Names
server.bundle.js  606 KiB       0  [emitted]  main
Entrypoint main = server.bundle.js
 [1] external "path" 42 bytes {0} [built]
 [5] external "fs" 42 bytes {0} [built]
 [7] external "buffer" 42 bytes {0} [built]
[10] external "stream" 42 bytes {0} [built]
[16] external "util" 42 bytes {0} [built]
[24] external "http" 42 bytes {0} [built]
[38] external "net" 42 bytes {0} [built]
[40] external "events" 42 bytes {0} [built]
[41] external "zlib" 42 bytes {0} [built]
[45] external "async_hooks" 42 bytes {0} [optional] [built]
[57] external "querystring" 42 bytes {0} [built]
[58] external "url" 42 bytes {0} [built]
[65] external "crypto" 42 bytes {0} [built]
[69] ./server.js 990 bytes {0} [built]
[80] external "tty" 42 bytes {0} [built]
    + 145 hidden modules

WARNING in ./node_modules/express/lib/view.js 81:13-25
Critical dependency: the request of a dependency is an expression
 @ ./node_modules/express/lib/application.js
 @ ./node_modules/express/lib/express.js
 @ ./node_modules/express/index.js
 @ ./server.js

sania270804@ubuntu:~/IdeaProjects/react-nodejs-example/api$
```

In the same api directory, build the API artifact.



View the generated bundle.

Task 6: Build a Docker Image for the Node.js Project

```
sania270804@ubuntu:~/IdeaProjects/node-project$ docker --version
Docker version 29.9.2, build a82e528

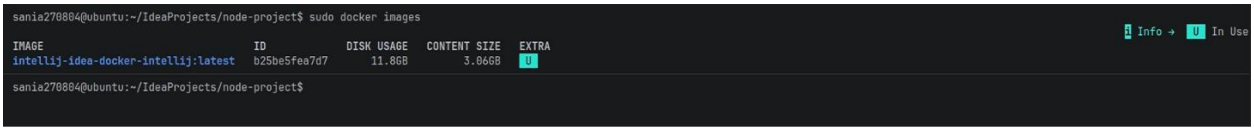
sania270804@ubuntu:~/IdeaProjects/node-project$
```

Check that the Docker command is installed.

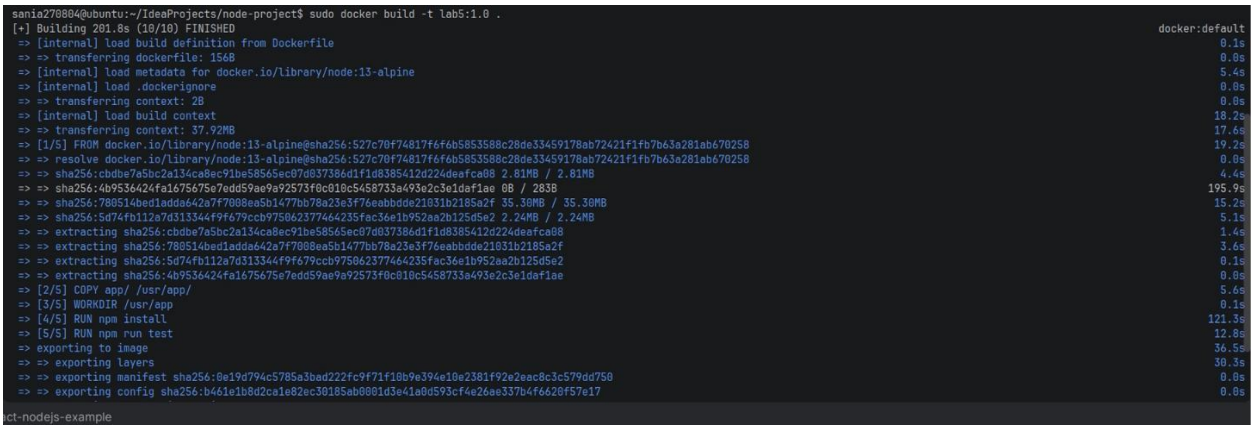
```
sania270804@ubuntu:~/IdeaProjects/node-project$ ls
Dockerfile  Jenkinsfile  README.md  app

sania270804@ubuntu:~/IdeaProjects/node-project$
```

List the repository files and locate Dockerfile.



View the existing Docker images before building.



Build the application image.

```
sania270804@ubuntu:~/IdeaProjects/node-project$ sudo docker images

IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
intellij-idea-docker-intellij:latest b25be5fea7d7       11.8GB         3.06GB   U
lab5:1.0                             7f6559bd3775        417MB          85.1MB

sania270804@ubuntu:~/IdeaProjects/node-project$
```

List the images again.